Fiscal Year:	FY 2021	**Task Last Updated:**	FY 07/02/2021
PI Name:	Hasenstein, Karl Ph.D.		
Project Title:	Assessment of Nutritional Value and Growth Parameters of Space-Grown Plants		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Bioregenerative Life Support		
PI Email:	hasenstein@louisiana.edu	**Fax:**	FY 337-482-5834
PI Organization Type:	UNIVERSITY	**Phone:**	337-482-6750
Organization Name:	University of Louisiana - Lafayette		
PI Address 1:	PO Box 42451		
PI Address 2:	300 E. St. Mary Boulevard		
PI Web Page:	http://www.ucs.louisiana.edu/		
City:	Lafayette	**State:**	LA
Zip Code:	70504-2451	**Congressional District:**	3
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	Space Biology Unsolicited
Start Date:	08/01/2017	**End Date:**	02/28/2022
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA KSC
Contact Monitor:	Levine, Howard	**Contact Phone:**	321-861-3502
Contact Email:	howard.g.levine@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date changed to 2/28/2022 per NSSC information (Ed., 7/8/21) NOTE: End date changed to 7/31/2021 per NSSC information (Ed., 5/4/2020)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	80NSSC17K0344		
Performance Goal No.:			
Performance Goal Text:			

Task Description:

The goal of the proposed work is to determine the effects of space flight on growth, performance (productivity and time to maturation), metabolism, and genetic changes in radish plants (Raphanus sativus). Plants will be grown in the ADVANCED PLANT HABITAT (APH) hardware on the International Space Station (ISS) and in comparable experiments in ground controls. The research conforms to NASA's top future goal of long-term space exploration and terraforming other planets and satellites. These goals depend on nutritionally valuable food crops that can be reliably grown in an environment that is not only defined by altered gravity or weightless environment but also different atmospheric conditions. In contrast to studying direct effects of gravity on plants, our approach focuses on secondary, derived conditions such as altered convection, transpiration, watering conditions, CO2, other gaseous components, and LED illumination, which creates a characteristic energy (heat and light) gradient within the growth volume.
Radishes are good candidate plants to grow on the ISS because they mature quickly (harvest after = 28 days after planting), have been grown successfully on the ISS (Massa et al., 2013); genetic and metabolic information is readily available (Shen et al., 2013); and they are nutritionally desirable and provide a wealth of information that can thoroughly be analyzed by biochemical, physiological, and genetic approaches. Radishes are good sources of minerals (especially K, Ca, Fe, and Zn), complex carbohydrates, vitamin C, dietary fibers, folic acid (vitamin B9), anthocyanins, and glucosinolates, a class of secondary metabolites characteristic for Brassicaceae. In addition to their roots, certain varieties of radish are used for their leaves, siliques, and oil. The nutritional value and versatility of these vegetables makes them good candidates as a crop plant for astronauts. The small seed size (<4 mm), longevity of seeds (up to five years), small plant size, and the ability to consume the entire plant makes radishes ideally suited for space biology.

Previous studies have focused on the effect of gravity and shown that space conditions allow successful cultivation of Brassica species (Musgrave et al., 2000). However, the quality, biomass, sensitivity of growth to day length, and (constant) temperature are important parameters for this species that have yet to be studied, especially in light of results that seed development in space cultures was compromised (Kuang et al., 2005), especially when oxygen was limited (Kuang et al., 1998). Equally unknown is the effect of environmental conditions in spacecrafts on the performance of plants. Understandably, the emphasis on space biology focuses on the effect of weightlessness and gravity on plant orientation and mechanical properties (Hoson 1999). However, conditions that are the consequence of weightlessness, e.g., lack of convection or altered atmosphere, are equally significant but unknown.

Our proposal intends to measure the metabolic, physiological, and genetic responses of radishes grown on the ISS and identify the effect of the space environment on metabolite accumulation, flavor, enzyme activity, mineral uptake, and generation time. This knowledge will be essential for the transition from Earth-bound cultivation of plants to growth conditions in space; the research is important for the establishment of human colonies on the Moon, Mars, and farther outposts.

References

Hoson T (1999) Interaction of gravity with other environmental factors in growth and development: An introduction. In: Duke PJ, Barlow PW, Hoson T, Mogami Y (eds) Life Sciences: Microgravity Research I, pp 1971-1974

Kuang A, Crispi M, Musgrave ME (1998) Control of seed development in Arabidopsis thaliana by atmospheric oxygen. Plant Cell and Environment 21: 71-78

Kuang A, Popova A, McClure G, Musgrave ME (2005) Dynamics of storage reserve deposition during Brassica rapa L. pollen and seed development in microgravity. Int J Plant Sciences 166: 85-96

Massa G, Newsham G, Hummerick ME, Caro JL, Stutte GW, Morrow RC, Wheeler RM (2013) Preliminary Species and Media Selection for the VEGGIE Space Hardware. Grav Space Res 1: 95-106

Musgrave M, Kuang A, Xiao Y, Stout S, Bingham G, Briarty L, Levinskikh M, Sychev V, Podolski I (2000) Gravity independence of seed-to-seed cycling in Brassica rapa. Planta 210: 400-406

Shen D, Sun H, Huang M, Zheng Y, Li X, Fei Z (2013) RadishBase: A Database for Genomics and Genetics of Radish. Plant Cell Physiol 54: E3-+

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

The objective to grow plants under space conditions requires optimization of auxiliary parameters such as light intensity, quality, watering, substrate definition, fertilization that all have immediate implications for more efficient agricultural practices. The data that we will obtain examine much more than simple yield (mass) of the plants but include the mineral content, propensity to flower, flavor components, and water use. All attempts will not only optimize plant culture under elevated CO2, limited water, and controlled fertilizer provision, but also investigate the genetic responses of plants to mitigate space (or agricultural) stress. The overall result of the project will therefore benefit our generic understanding of factors that control plant growth and productivity in space and on Earth.

Space experiment:
The objective of the space flight experiment was the assessment of crop plant performance in space and to analyze the effect of weightlessness, elevated CO2, specific light conditions, and artificial substrate on plant growth and productivity.

The preparations for the flight experiments were completed and the prepared (seeded) science carrier launched on the Cygnus 14 flight on 10/2/20. The first experiment was initiated on November 3 and completed by 11/30/2020. The second grow-out started 12/1 and was completed 12/30/2020. Prior to return, samples were stored on the International Space Station (ISS) in the Minus Eighty-Degree Laboratory Freezer for ISS (MELFI) freezer. Sample return of the biological material from the first ISS grow-out occurred May 2 and samples were transported to and stored at the NASA Kennedy Space Center (KSC). The sample return was incomplete because some leaf material was stored in a separate bag that was inadvertently left on the ISS and hopefully will be returned in July. Available samples of the first grow-out returned to the lab on June 2, 2021. The analysis of samples has been initiated and will include genetic analyses, comparisons with a ground control experiment (performed at KSC from 12/17/20 to 1/13/21). This report includes two separate activities: the comparison of remote data from ISS 1 and 2 and the ground control performed at KSC. The KSC ground control will be compared with a (UL) lab control experiment that replicates the KSC ground control (conducted at 2500 ppm CO2) at normal (400 ppm) CO2 levels.

Task Progress:	Growth experiments performed in orbit (ISS1 and ISS2): Based on prior experience during the science verification test (SVT) and experiment verification test (EVT), the following parameters were chosen for the ISS experiments to examine growth of radish plants during a 27-day growth period in the Advanced Plant Habitat (APH). The CO2 concentration was set at 2500 ppm, reduced from the current 3500 ppm level on Station. Light intensity (Photosynthetically Active Radiation) was set at 500 µMol m-2s-1; the temperature settings fluctuated between 24°C during the light cycle and 20°C during the dark period. Temperature transition occurred over a one-hour period. The relative humidity (RH) was set at 80% during the nine days and then reduced to 65%. Diurnal fluctuations in RH values ranged between 65 and 75%. Although tissue collection and harvest were obviously performed by astronauts, the chamber was controlled remotely from the ground via the APH software. Plant cultivation: Each of the twenty growth positions received three seeds that were secured with water-soluble glue to floral foam. After eight days (11/11/20), the plants were thinned and only the most vigorous seedling per position was retained. Leaf samples were taken after two additional days (11/13/20); leaf samples were taken transferred to microfuge tubes and stored at -80°C. Additional leaf punch material was collected 17 and 24 days after planting (11/20 and 11/27). Plants in all twenty positions grew comparably to ground controls (see below). After 27 days, the plants were harvested by first removing the leaves as complete canopy and photographing the spread-out leaves to assess leaf area and appearance. For future analyses, the leaves were folded into aluminum foil pouches and stored at -80°C (MELFI). After leaf harvest, the radishes (i.e., bulb tissue) were excised from the arcillite substrate, transferred into separate plastic bags, and also stored in the MELFI. Part of the samples from the first grow-out were returned from the ISS on May 2nd and stored at KSC. These samples were transported in insulating boxes containing dry ice and delivered to University of Louisiana (UL) Lafayette on June 2nd. Of the 15 documented leaf samples, seven were returned. The remainder stayed on the ISS and are scheduled to be returned with SpaceX 22. APH performance and plant development: Because plant performance is intimately linked to the growth environment, the first part of the report compares the ground control with the two grow-outs on orbit. Although initially two ground controls were planned, a computer update interfered with the first ground control at KSC and caused power outages for three consecutive days. A second ground control proceeded without power problems but still showed substantial differences between ISS conditions. The differences pertained to the lighting system (520 µM m-2s-1 on the ISS versus 380 µM m-2s-1 during the first 24 days of the ground control), the relative humidity (RH) of the chamber (initially 80% in both cases but after the first week 65- 70 % for the ISS unit and 80% during the third week for the ground control), and the CO2 concentration (2500 ppm for the ground control but lower and highly fluctuating values (1800 – 2300 ppm) for the first grow-out on the ISS). The second grow-out on the ISS had only 2000 ppm CO2 during the entire growth period with much lower variability. The water delivery and moisture levels for both ISS experiments were remarkably similar but – as expected – differed from the ground control, which showed a strong top/bottom gradient in the moisture readings that was absent in the space experiments. The readings of all moisture sensors (two per quadrant, one at the higher position, ca. 10 mm below the surface and a lower sensor, about 10 mm above the bottom of each science carrier quadrant) showed strong daily fluctuations that controlled water delivery. For the ISS experiments, the upper sensors values were reduced but the difference between top and bottom sensors was lower and probably the result of rather high air movement (fans were set at 0.9 m/s) that enhanced evaporation of the top arcillite layer. Higher variability of the sensor readings coincided with larger water demand. The temperature profile for ground control and ISS experiments was comparable with high values during the photoperiod of 24°C and 20°C during dark periods. The evaluation of mineral content, qPCR analysis in nitrogen, and glucosinolate measurements have been initiated. In contrast, RNA-seq evaluation is on hold until the material from the second ISS experiment will become available. For cost-saving reasons, all samples will have to be processed at the same time. Plant productivity: The leaf area per quadrant for the first ISS experiment was 938 cm2, which compares well with the data from the EVT (1033 cm2) and ? EVT (818 cm2). Weight determinations for the radish bulbs is still missing as samples need to not be thawed before processing for genetic analyses. Those will be performed with samples from the second ISS grow-out. Therefore, these data are not yet available. Lab work 1 - Carbohydrate analyses: In preparation for the analyses of space samples, we established the protocol for carbohydrate analysis and quantification of glucosinolates. Carbohydrate analyses (based on High Performance Liquid Chromatography (HPLC) measurements with an Hypersil-gold amino, 250 x 4.6 mm column and 80% acetonitrile as isocratic mobile phase) showed that the prominent sugars of bulbs are fructose and sucrose. Examined sugars included 3-O-Methyl glucopyranose, Arabinose, Fructose, Fucose, Galactosamine, Galactose, Glucose, Lactose, Maltose, Mannitol, Mannose, Myo-inositol, Psicose, Rhamnose, Sorbitol, Sucrose, Trehalose, and Xylose. 2 – Quantification of glucosinolates. We developed a new method based on sodium palladate for the direct quantification of glucosinolates. This method is faster, more sensitive, and less error prone (does not require strong anion exchange chromatography) than the originally employed, potassium ferricyanide based method (Gallaher et al. 2012). Sodium palladate calibration curve (y = 0.0058 x (=µg sinigrin) -0.0046) has a correlation coefficient of 0.9987 and is linear between 0 to > 40 µg sinigrin. Reference: Gallaher CM, Gallaher DD, Peterson S (2012) Development and Validation of a Spectrophotometric Method for Quantification of Total Glucosinolates in Cruciferous Vegetables. J Agric Food Chem 60: 1358-1362.
Bibliography Type:	Description: (Last Updated: 01/20/2026)

Articles in Peer-reviewed Journals	Pesacreta TC, Acharya A, Hasenstein KH. "Endogenous nutrients are concentrated in specific tissues in the Zea mays seedling." Protoplasma. 2021 Jul;258(4):863-78. Epub 2021 Feb 13. https://doi.org/10.1007/s00709-021-01606-4 ; PMID: 33582844 [Correction in: Protoplasma. 2021 Mar 2] , Jul-2021
Articles in Peer-reviewed Journals	Kordyum E, Hasenstein KH. "Plant biology for space exploration – Building on the past, preparing for the future." Life Sci Space Res (Amst). 2021 May;29:1-7. https://doi.org/10.1016/j.lssr.2021.01.003 ; PMID: 33888282 , May-2021
Articles in Peer-reviewed Journals	John S, Abou-Issa F, Hasenstein KH. "Space flight cultivation for radish (Raphanus sativus) in the Advanced Plant Habitat." Gravit Space Res. 2021 Aug 22;9(1):121-32. https://doi.org/10.2478/gsr-2021-0010 , Aug-2021